Derivative Applications and Modeling
Answer Key

AP Calculus AB/BC   Applications of Derivatives

Quick Answers

1. $3x^2 - 12x + 9$
2. 1, 3
3. 200π
4. 15, 450
5. 0, 2
6. $-\frac{5}{3}$
7. 2, 6, 128
8. -1, 3, 5, -22
9. $\frac{1}{\pi}$
10. $\sqrt{3}$, $-\sqrt{3}$

Curriculum

Level: AP Calculus AB/BC   Applications of Derivatives
CHA-3.D – *problems 3, 6, 9*
FUN-3.A–FUN-3.C – *problem 1*
FUN-4.B–FUN-4.C – *problems 2, 4, 5, 7, 8, 10*
Difficulty 1–5 of 5 across 15 tagged checks.

Common wrong answers

3. *If they got 314.16: differentiated the volume formula but left off the dr/dt factor (no chain rule)*
9. *If they got 0.0796: used the full radius instead of r = h/2, so the cross-section area came out four times too large*

1. Differentiate the cubic.

Step 1. Every term is a power of x, so the power rule alone is enough — no product or chain rule is hiding here.
Step 2. Term by term: $3x^2$ from x^3, $-12x$ from $-6x^2$, 9 from $9x$, and 0 from the constant 2.
Step 3. $\boxed{f'(x) = 3x^2 - 12x + 9}$.

2. Critical numbers and increasing intervals.

Step 1. f is a polynomial, so f' exists everywhere and the only critical numbers are the zeros of f'.
Step 2. $3x^2 - 12x + 9 = 3(x^2 - 4x + 3) = 3(x-1)(x-3) = 0$.
Step 3. The critical numbers are $\boxed{x = 1 \text{ and } x = 3}$.
Step 4. Sign of $f' = 3(x-1)(x-3)$: both factors negative for $x < 1$ (product positive), signs opposite on $1 < x < 3$ (negative), both positive for $x > 3$. So f increases on $(-\infty, 1)$ and on $(3, \infty)$ and decreases on $(1, 3)$.
Grading note: the sign chart, or an explicit test value in each interval, is the justification. "f' is a parabola opening upward" with no values shown is not.

3. Expanding balloon (related rates).

Step 1. Radius and volume both change with time, so differentiate the relationship with respect to t first and substitute afterwards.
Step 2. From $V = \frac{4}{3}\pi r^3$, the chain rule gives $\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$.
Step 3. At the required instant $r = 5$ and $\frac{dr}{dt} = 2$, so $\frac{dV}{dt} = 4\pi(25)(2) = \boxed{200\pi}$ cubic centimetres per second.

Common wrong answer: 314.16 — the $\frac{dr}{dt}$ factor was dropped, which computes the surface area rather than a rate.

4. Fenced garden (optimization).

(a) Step 1. $A(x) = x(60 - 2x) = 60x - 2x^2$ is already a function of one variable on the physical domain $0 < x < 30$, so differentiate it directly.
Step 2. $A'(x) = 60 - 4x = 0$ gives $4x = 60$.
Step 3. $\boxed{x = 15}$ metres.
Step 4. Justification: $A''(x) = -4 < 0$ everywhere, so the graph of A is concave down and this single critical point is the absolute maximum on the interval.
(b) Step 5. The other side measures $60 - 2(15) = 30$ metres, so $A(15) = 15 \cdot 30$.
Step 6. The maximum area is $\boxed{450}$ square metres.

5. Where $f'' = 0$, and which of those are inflection points.

Step 1. An inflection point needs a change of concavity, so start from the second derivative and then test it for a sign change — $f'' = 0$ alone is not enough.
Step 2. $f(x) = x^4 - 4x^3$ gives $f'(x) = 4x^3 - 12x^2$ and $f''(x) = 12x^2 - 24x = 12x(x - 2)$.
Step 3. $f''(x) = 0$ at $\boxed{x = 0 \text{ and } x = 2}$.
Step 4. Sign of f'': positive for $x < 0$, negative on $0 < x < 2$, positive again for $x > 2$. The concavity changes at both, so both are inflection points, at $(0, 0)$ and $(2, -16)$.
Teaching note: contrast this with $g(x) = x^4$, where $g''(0) = 0$ but g'' never changes sign, so the origin is not an inflection point. That is why Step 4 exists.

6. Sliding ladder (related rates).

Step 1. Let x be the distance from wall to foot and y the height of the top. The ladder length is constant, so $x^2 + y^2 = 169$ holds at every instant and can be differentiated in t.
Step 2. At the instant asked about, $y = \sqrt{169 - 25} = 12$. Compute y from the relation — it is not a given.
Step 3. Differentiating: $2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$, so $\frac{dy}{dt} = -\frac{x}{y} \cdot \frac{dx}{dt} = -\frac{5}{12}(4)$.
Step 4. $\frac{dy}{dt} = \boxed{-\frac{5}{3} \text{ ft/s}}$.
Reading the sign: negative means y is decreasing — the top is sliding *down* at $\frac{5}{3}$ feet per second. A positive answer here is a sign error, not a different convention.

7. Open box (optimization with a domain check).

(a) Step 1. Expand before differentiating: $V(x) = x(12 - 2x)^2 = 4x^3 - 48x^2 + 144x$.
Step 2. $V'(x) = 12x^2 - 96x + 144 = 12(x-2)(x-6) = 0$, so the roots are $\boxed{x = 2 \text{ and } x = 6}$.
Step 3. The domain rejects $x = 6$: cutting 6 inches from each corner of a 12-inch sheet removes the whole width, and $V(6) = 0$ is a degenerate box, not a maximum. Only $x = 2$ lies in $0 < x < 6$.
(b) Step 4. $V(2) = 2(12 - 4)^2 = 2 \cdot 64$.
Step 5. The maximum volume is $\boxed{128}$ cubic inches.
Step 6. Justification: $V' = 12(x - 2)(x - 6)$ is positive on $(0, 2)$ and negative on $(2, 6)$, so V rises then falls and $x = 2$ is the absolute maximum on the physical domain.
Grading note: an answer of $x = 6$ is the whole point of the problem — the algebra is right and the model is not.

8. Absolute extrema on a closed interval.

(a) Step 1. $f'(x) = 3x^2 - 6x - 9 = 3(x^2 - 2x - 3) = 3(x - 3)(x + 1)$.
Step 2. The critical numbers are $\boxed{x = -1 \text{ and } x = 3}$. Only $x = 3$ lies in $[0, 4]$; $x = -1$ is outside and plays no part.
(b) Step 3. On a closed interval the candidates are the interior critical numbers and both endpoints — the Extreme Value Theorem guarantees the extrema exist and that this short list contains them.
Step 4. $f(0) = 5$; $\quad f(3) = 27 - 27 - 27 + 5 = -22$; $\quad f(4) = 64 - 48 - 36 + 5 = -15$.
Step 5. The absolute maximum is $\boxed{5}$, at $x = 0$, and the absolute minimum is $\boxed{-22}$, at $x = 3$.
Listen for: a student who stops at the critical number and reports -22 as "the" answer. On a closed interval the endpoints are candidates every time, and here the maximum is at one of them.

9. Filling cone (related rates).

Step 1. The volume formula has two changing variables, r and h. The similar-triangles relation $r = h/2$ removes one *before* differentiating, which is what makes the problem solvable.
Step 2. $V = \dfrac{1}{3}\pi r^2 h = \dfrac{1}{3}\pi \left(\dfrac{h}{2}\right)^2 h = \dfrac{\pi h^3}{12}$.
Step 3. Differentiate in t: $\dfrac{dV}{dt} = \dfrac{\pi h^2}{4} \cdot \dfrac{dh}{dt}$, so $\dfrac{dh}{dt} = \dfrac{4}{\pi h^2} \cdot \dfrac{dV}{dt}$.
Step 4. With $\dfrac{dV}{dt} = 9$ and $h = 6$: $\dfrac{dh}{dt} = \dfrac{4(9)}{\pi(36)} = \boxed{\dfrac{1}{\pi} \text{ ft/min}}$ (about 0.32 feet per minute).
Common wrong answer: about 0.08 — the substitution $r = h/2$ was skipped, so the cross-section was taken four times too large.

10. Synthesis: inscribed rectangle.

(a) Step 1. $A(x) = 2x(9 - x^2) = 18x - 2x^3$, so $A'(x) = 18 - 6x^2$.

Step 2. $18 - 6x^2 = 0$ gives $x^2 = 3$, so the roots are $\boxed{x = \sqrt{3}}$ and $\boxed{x = -\sqrt{3}}$. Only the positive root lies in $[0, 3]$; the negative root describes the same rectangle reflected and is outside the stated domain.

(b) Step 3. *(Open response — flagged for manual review; read the student's argument.)* A full-credit justification does all of this:

- names $[0, 3]$ as a closed, bounded interval and A as continuous on it, so the Extreme Value Theorem guarantees an absolute maximum exists;
- compares the only three candidates — $A(0) = 0$, $A(3) = 0$, and $A(\sqrt{3}) = 2\sqrt{3}(9-3) = 12\sqrt{3} \approx 20.8$ — or else gives the sign of A' on each side of $\sqrt{3}$ (positive then negative);
- explains that a table of sampled values can only report the largest value *it happened to sample*. Between any two sampled inputs the function is unconstrained, so a table can suggest a maximum but never establish one; the derivative argument rules out every unsampled input at once.

Grading note: accept either the candidate-comparison or the first-derivative-sign argument, but require the closed-interval or sign-change reasoning to be stated. "$A(\sqrt{3})$ is bigger than the values I tried" is exactly the answer part (b) is asking the student to improve on.